Grade Eight Science
Body Systems Design Lab Information and Completion Schedule

WHAT WE WILL DO:

COLLABORATIVE:
- Work collaboratively to come up with an idea and design an experiment based on any body system where we can collect quantitative data
- Work collaboratively to collect the data

INDEPENDENT:
- Write our own lab report using the idea and data that we collected collaboratively that is based on MYP Criterion B and C (posted below)

MYP Command Terms that are used on this assessment:

State: Give a specific name, value, or other brief answer without explanation of calculator	**Outline:** Give a brief account or summary
Discuss: Offer a considered and balanced review that includes a range of arguments, factors or hypothesis. Opinions or conclusions must be presented clearly and supported by appropriate evidence.	**Describe:** Give a detailed account or picture of a situation, event, pattern, or process
Explain: Give a detailed account including reasons and causes	**Identify:** Provide an answer from a number of possibilities. Recognize and state briefly a distinguishing fact or feature.
Present: Offer for display, observation, examination, or consideration	**Organize:** Put ideas and information into a proper or systematic order
Interpret: Use knowledge and understanding to recognize trends and draw conclusions from given information	**Select:** Choose from a list or group (or select appropriate materials for independent work)

MYP ASSESSMENT CRITERIA:

- The design lab will be graded on two MYP assessment Criteria: Criterion B (Inquiring and Designing) and Criterion C (Processing and Evaluating) (Posted below)
- You will receive two grades for the lab: for for Criterion B and one for Criterion C

RUBRIC 1: CRITERION B - INQUIRING AND DESIGNING

Achievement Level	Task Specific Level Descriptor
1-2	The student: • **states** a research question with **limited success** • **states** a testable hypothesis • **states** variables • designs a **method with limited success**
3-4	The student: • **states** a research question • **outlines** a testable hypothesis using **scientific reasoning** • **outlines** some variables and their manipulation correctly • designs a **safe method** that **outlines** how some **relevant** data will be collected and **selects** some **materials**
5-6	The student: • **outlines** a valid research question with minimal error • **outlines and explains** a working hypothesis using scientific knowledge • **outlines** most variables and their manipulation correctly • designs a **complete, safe method** that **outlines** how **sufficient, relevant** data will be collected and selects **appropriate materials** with minimal error
7-8	The student: • accurately **describes** a valid research question • correctly **outlines and explains** a hypothesis **using correct scientific reasoning** • **Describes correct scientific reasoning** to assign and manipulate variables • designs a **logical, complete, safe method** that **describes** how data will be collected and selects **appropriate materials**

RUBRIC 2: CRITERION C - PROCESSING AND EVALUATING

<table>
<tr><th>Achievement Level</th><th>Task Specific Level Descriptor</th></tr>
<tr><td>1-2</td><td>The student:
collects quantitative data and presents data into a table and graph with limited success
accurately interprets data with limited or no explanation
states if the hypothesis is correct or incorrect with limited or no reference to the results
states errors with limited reference to the method
states limited ways to improve the experiment</td></tr>
<tr><td>3-4</td><td>The student:
correctly collects quantitative data and organizes and presents data into a table and graph with some error
accurately interprets data and describes the results by filling in the explanation section with some error
states if the hypothesis is correct or incorrect with some reference to the results
states valid errors based on the method of the experiment with some explanation
states two ways to improve the experiment with some explanation</td></tr>
<tr><td>5-6</td><td>The student:
correctly collects quantitative data and correctly organizes and presents raw and average data into a table and graph with minimal error
accurately interprets data and correctly describes the results by filling in the explanation section with scientific reasoning with minimal error
outlines the validity of the hypothesis based on appropriate evidence from the experiment
outlines, based on a balanced review of the method, errors that can impact the experiment with minimal error
outlines ways to improve the experiment with detail of what changes need to be made</td></tr>
<tr><td>7-8</td><td>The student:
correctly collects quantitative raw data, transforms it into average data, and correctly organizes and presents data into a table and graph
accurately interprets data and correctly describes the results by filling in the explanation section with scientific reasoning
discusses the validity of the hypothesis based on appropriate evidence from the experiment
discusses, based on a balanced review of the method and using appropriate evidence, errors that can impact the experiment
describes ways to improve the experiment by explaining how the changes would positively impact the method and results</td></tr>
</table>

DESIGN LAB COMPLETION SCHEDULE

Class	**What's Happening in Each Class**
	CRITERION B: INQUIRING AND DESIGNING
1 and 2	• The lab is introduced and partners assigned • Students choose a topic about any body system that allows them to collect quantitative data **(Collaborative)** • Write background information for your experiment that **describes** the problem or question to be tested by scientific investigation and **outlines** and **explain** a testable hypothesis using correct scientific reasoning **(Individual)**
3	• **Outline** and **explain** the hypothesis • **Describes** the control, independent and dependent variables and their manipulation • Select appropriate materials and write a materials list **(Individual)**
4	• Design a logical, complete, safe method (procedure) that **describes** how the data will be collected **(Individual)**
5	• Review the research question, hypothesis, variables, materials, and method **(Individual)** • **Submit this section of the lab on Google Classroom (Individual)**
CRITERION C	**PROCESSING AND EVALUATING**
6	• **If you have not submitted the work for "Criterion B," submit it now** • Data Collection - we will run three trials of our experiment **(Collaborative)**
7 and 8	• Transform the collected data into a table with table number and title and units **(Individual)** • Transform the average data into a graph with Figure number and title and all axis labels and units **(Individual)**
9	• Finish up any unfinished items from classes 5 and 6 • Write a summary of trends in the data collected by accurately **interpreting** the data and **describing** the results using scientific reasoning **(Individual)**

10/11	• Write a conclusion that **discusses** the validity of the hypothesis (proven correct or incorrect) **(Individual)** • **Discusses** two errors, based on a review of the method and on appropriate evidence, that can impact the experiment **(Individual)** • **Describes** ways to improve the experiment by explaining how changes would positively impact the method and results **(Individual)**
12	• Overflow day • This is a day to finish up any unfinished sections of the lab **(Individual)** • This is a day to proof read and edit your lab • **Submit your lab** on Google Classroom at the end of the class **(Individual)**